

End of Topic Test – Chapter 4: Critical Analysis (Test 2) — Mark Scheme

Total: 51 marks

This is a critical-analysis paper: most answers are extended-response. Marks below are indicative points that should be credited (B marks, awarded independently); any other well-reasoned, relevant point of equal quality should be credited equivalently.

1. Identifying evidence, reasoning and conclusions

(a) B2 (or B1+B1) the evidence is the survey of 350 local drivers finding 74% said they would reduce their speed if cameras were installed. [2]

(b) B2 (or B1+B1) the reasoning: because most drivers say they would reduce speed with cameras present, installing cameras is inferred to reduce actual driving speeds, which is then linked to fewer accidents. [2]

(c) B2 (or B1+B1) conclusion: installing speed cameras will reduce the number of accidents on the ring road. [2]

(d) B1 sample representativeness: 350 drivers (self-reported survey responses) may not represent all road users (e.g. pedestrians, cyclists, other vehicles affected by accidents). B1 assumption: stated intention (“would reduce speed”) does not guarantee actual driving behaviour changes once cameras are installed. B1 logical leap: even if drivers slow down, this does not automatically prove accidents will fall (does not account for other causes of accidents, e.g. road layout, weather, driver distraction). [3]

2. Evaluating claims about breakfast clubs and attainment

(a) B1 the DfE statement shows a large-scale rollout (2,700+ schools) explicitly aimed at helping pupils arrive ready to learn and improving attendance/concentration – factors linked to attainment, the same outcome the newspaper claims. B1 this gives some authoritative backing/scale to the idea that breakfast provision is linked to improved exam results. B1 caveat: it shows aims/intent and scale of rollout, not proof that exam results actually improve as a result. [3]

(b) B1 the NPRT review states evidence on attainment impact is “mixed”. B1 it reports only “small improvements” in concentration/attendance, not exam grades specifically. B1 it states effects on exam grades are “inconsistent and often not statistically significant”, directly contradicting the newspaper’s stronger claim of a “significant” improvement. B1 conclusion: this weakens confidence that breakfast clubs alone cause meaningful improvement in exam results. [4]

(c) B1 balances both sources: the DfE quotation shows scale/intent (attendance, concentration) but no proven exam outcome; the NPRT quotation shows attainment evidence is inconsistent and often not significant. B1 concludes the newspaper’s claim of a “significant” improvement in exam results is an overstatement not fully supported by the evidence provided. B1 quality/clarity of the overall evaluation, referencing both sources explicitly. [3]

3. Interpreting claims about recycling trends

(a) B1 the headline is based on a single year’s fall (46% to 43%), a short-term fluctuation compared with the multi-decade rising trend shown in the chart. B1 a one-year dip does not by itself indicate a “collapsing” scheme – recycling rates can fluctuate year to year for many reasons (e.g. contamination rules, weather, contractor changes) without a genuine reversal in the underlying trend. B1 the chart shows recycling has more than tripled since 2005 (roughly 15% to 46%), so even with the recent dip, current levels remain far above historic levels – “collapsing” language exaggerates the scale of the change. [3]

(b) B1 the officer's quotation shows recycling has more than tripled since 2005, a substantial long-term improvement. B1 this suggests the one-year fall is small relative to the much larger long-term rise, so does not constitute a genuine reversal or "collapse". B1 conclusion: describing the scheme as "collapsing" is not well supported when the full time series is considered. [3]

(c) B1 a reasonable suggestion (e.g. recycling rates for the following year to see if the fall continues or is a one-off blip, a breakdown of which materials fell, or details of any changes to collection rules/contractors during 2024). B2 developed justification of why this would help judge the claim (e.g. one year's data alone cannot distinguish a genuine downward trend from a temporary dip). [3]

4. Sampling and bias

(a) B2 (or B1+B1) convenience sampling (also acceptable: opportunity/non-random sampling) – the manager surveyed the members who happened to attend one particular class rather than selecting randomly from all members. [2]

(b) B1 members attending an early-morning spin class may be more health/fitness-focused or have different attitudes to price than the wider membership (attitude/behaviour-related bias). B1 all 25 are from one class at one time of day, so may share similar characteristics (e.g. availability, income, age), reducing representativeness of the whole membership. B1 (third, credit if two are not fully developed) the small sample size (25) limits how confidently results generalise to all members. [3]

(c) B1 a more appropriate method, e.g. stratified random sampling across all membership types/time slots (proportional to numbers in each group). B1 how it would be implemented (random selection of members within each stratum using the gym's membership database). B1 why it is better (ensures members with different attendance patterns/membership types are represented proportionally, reducing bias and improving generalisability to the whole membership). [3]

5. Correlation and causation

(a) B2 (or B1+B1) correlation means there is a statistical association between two variables (ice cream sales and drowning incidents) – as one tends to increase, so does the other – but this alone does not establish that one causes the other. [2]

(b) B1 there is a confounding (third) variable – hot weather – which is likely to increase both ice cream sales and the number of people swimming (and therefore at risk of drowning). B1 the key point is that both variables are driven independently by a common cause rather than one causing the other. B1 correlation alone cannot distinguish between a genuine causal link and a shared underlying cause. [3]

(c) B2 (or B1+B1) a suitable suggestion, e.g. data on temperature/weather alongside sales and drowning figures, or data on the number of people swimming/visiting beaches each month, to see whether this better explains the pattern than ice cream sales alone. [2]

6. Reasoning flaws: meal kit boxes and food waste

(a) B1 the article treats "pre-measured ingredients" as equivalent to "low-waste", ignoring the packaging waste (plastic wrapping, ice packs, cardboard) generated by each box. B1 subscriber growth figures are used as if they were evidence of a low-waste outcome, but growth in usage does not itself prove reduced waste. B1 vague/unquantified claims ("fuelled by growing concern about food waste") are presented as supporting evidence rather than being substantiated. [3]

(b) B1 alternative explanation 1, e.g. growth could be driven by convenience/time-saving rather than waste concerns. B1 alternative explanation 2, e.g. effective marketing/advertising spend,

subscription discounts, or a general rise in home cooking. B1 explanation clearly linked back to why this would affect subscriber numbers independent of food-waste concerns. [3]

(c) B1 the quotation shows that a meal kit box's true environmental impact depends on packaging, delivery distance and whether every meal is actually cooked – factors the newspaper's claim ignores. B1 this challenges the newspaper's implied blanket conclusion that meal kits solve the food waste problem, showing the real picture is more nuanced and could still generate significant (packaging) waste. [2]